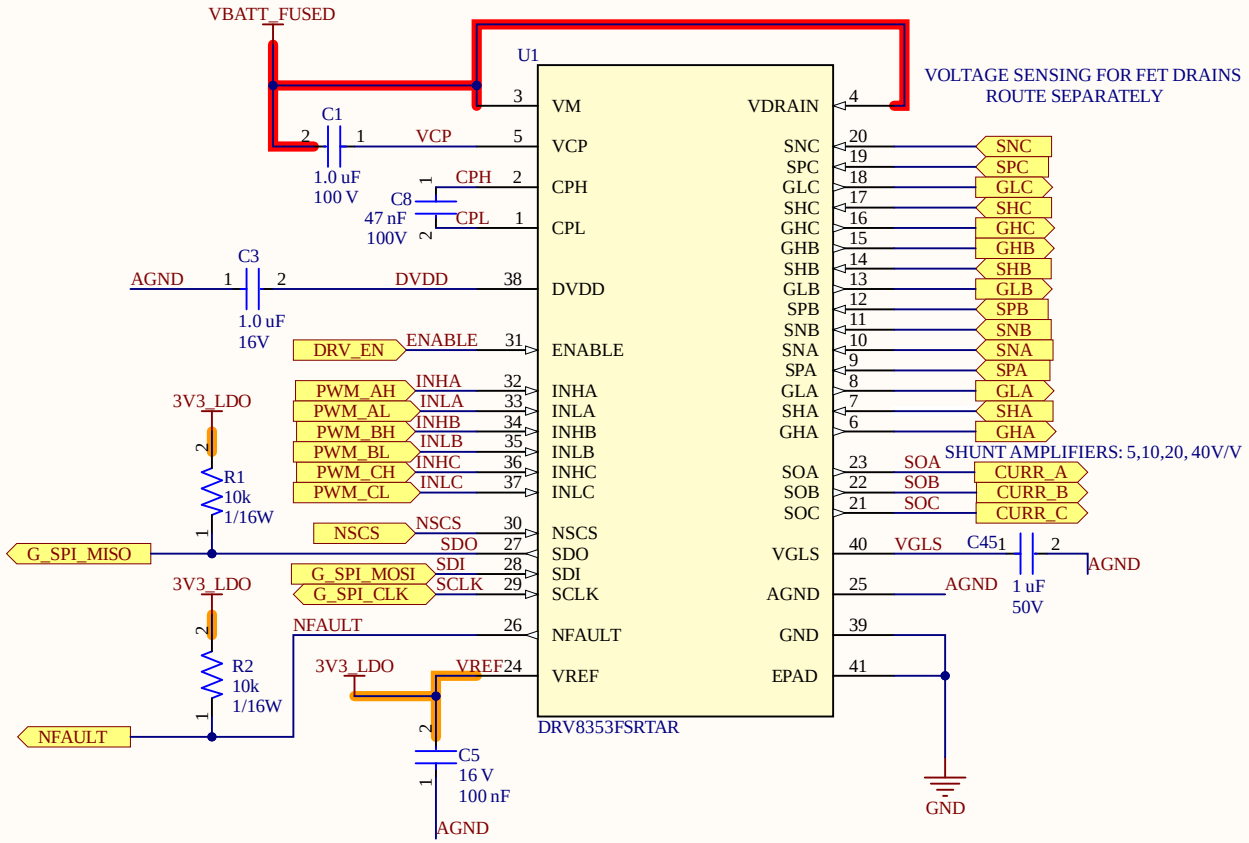
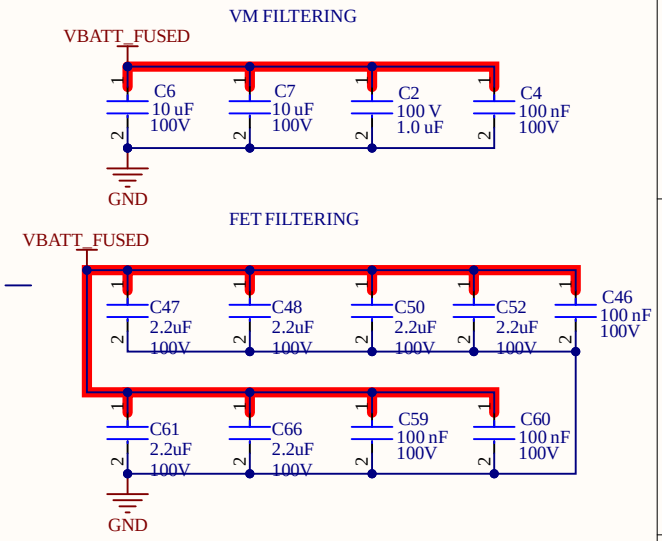
GATE DRIVER

SPx and SNx are shunt amplifier inputs
SOx are shunt amplifier outputs
GHx is high side gate driver
GLx is low side gate driver

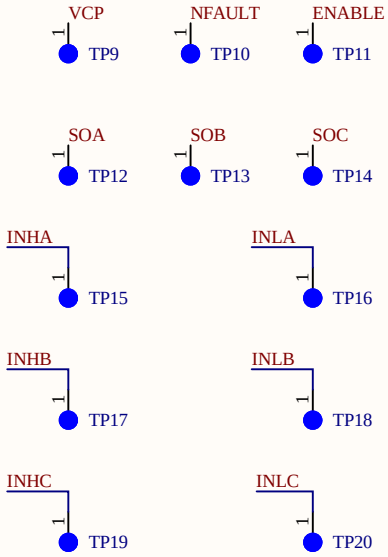


9.34A CONTINUOUS RATED AT 24V
WITH 48V BATTERY, MAX CONTINUOUS AMPERAGE IS 0.5*9.34 = 4.7A DRAWN FROM BATTERY, MOTOR EFFECTIVELY SEES 24V

EXTRA FILTERING

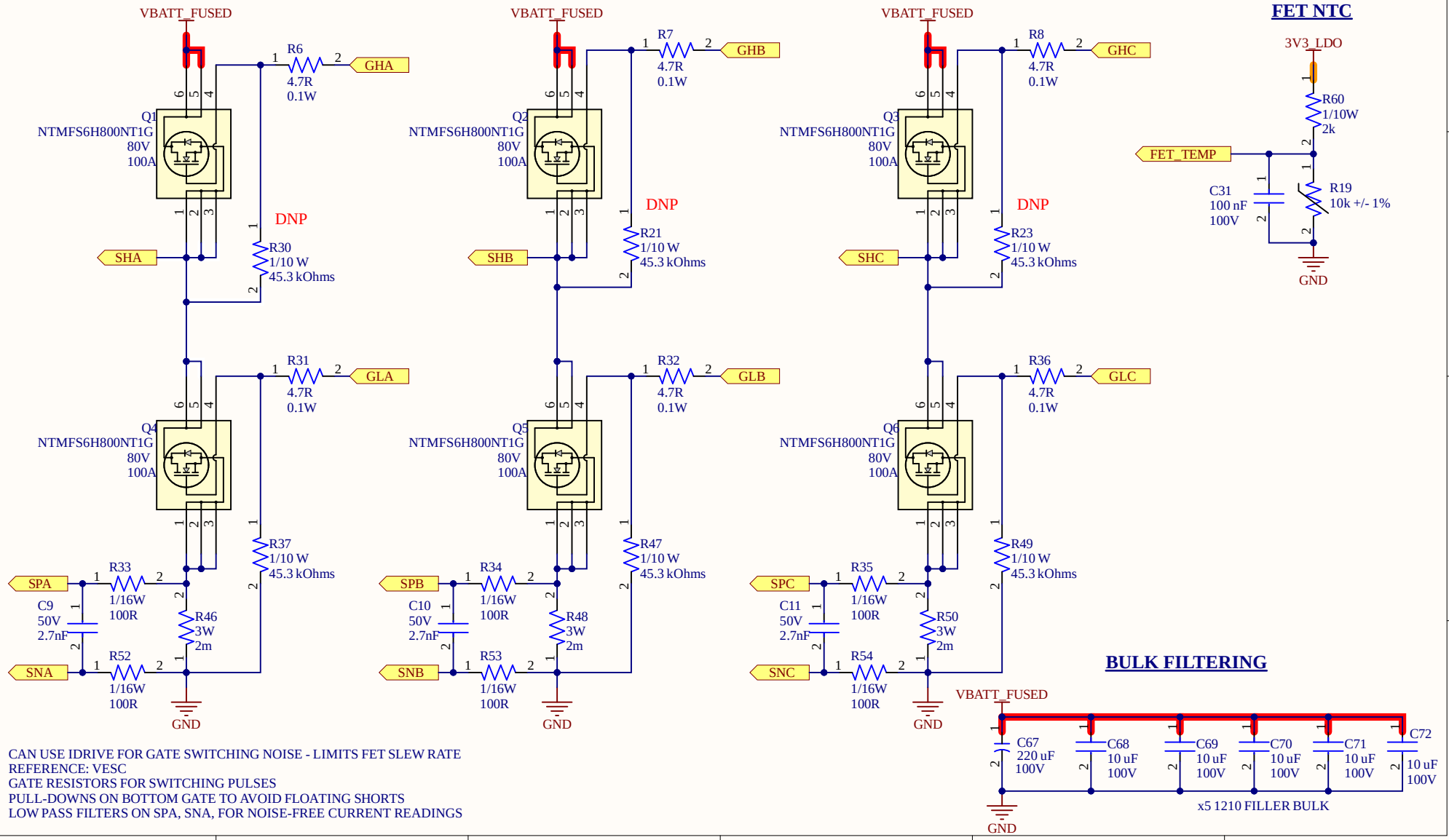


TEST POINTS



H-BRIDGE

PUT BIG COPPER PLANES UNDERNEATH FET, SEE DATASHEET
START AT GATE WITH 0OHM RESISTORS

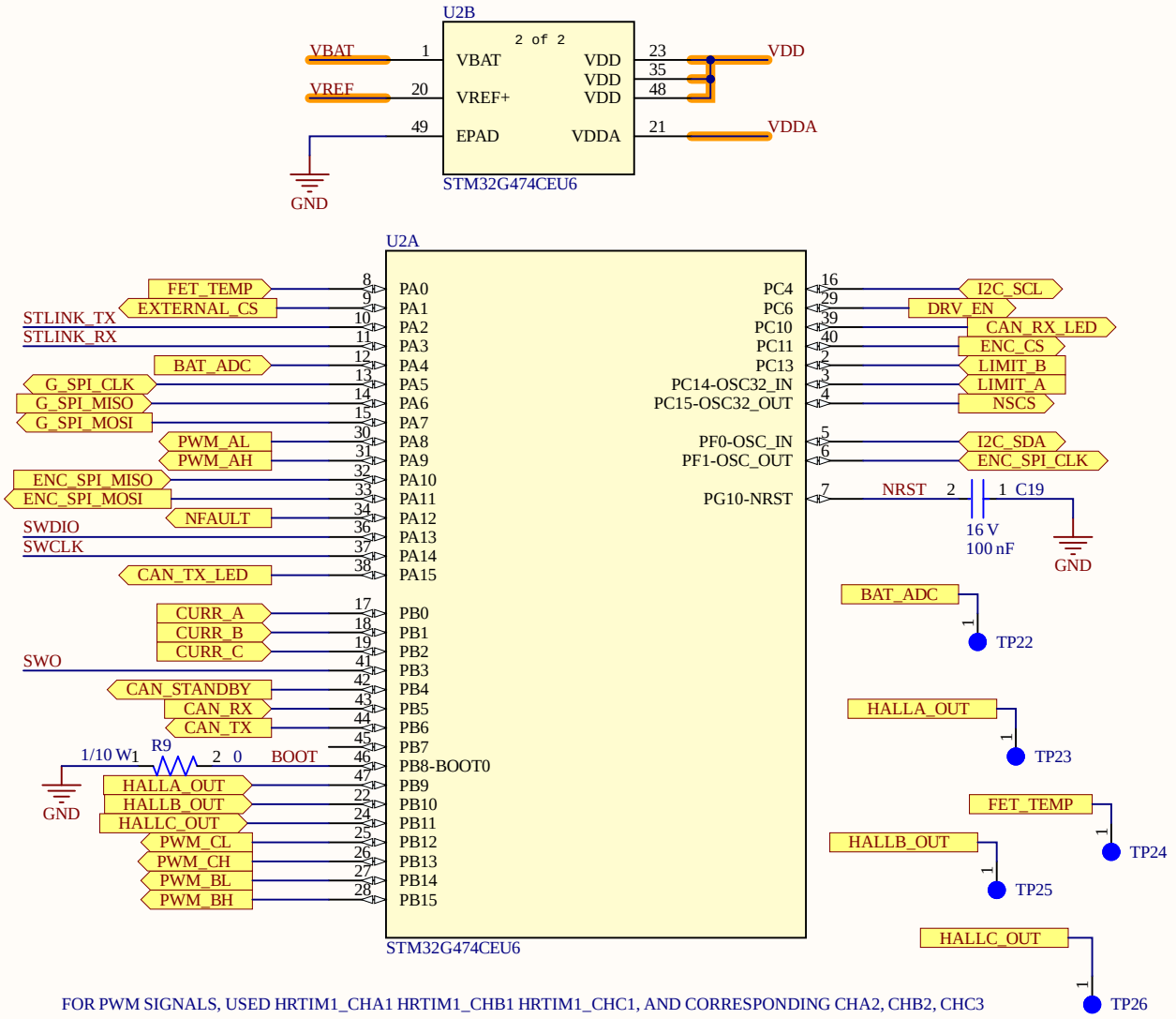


FET NTC

BULK FILTERING

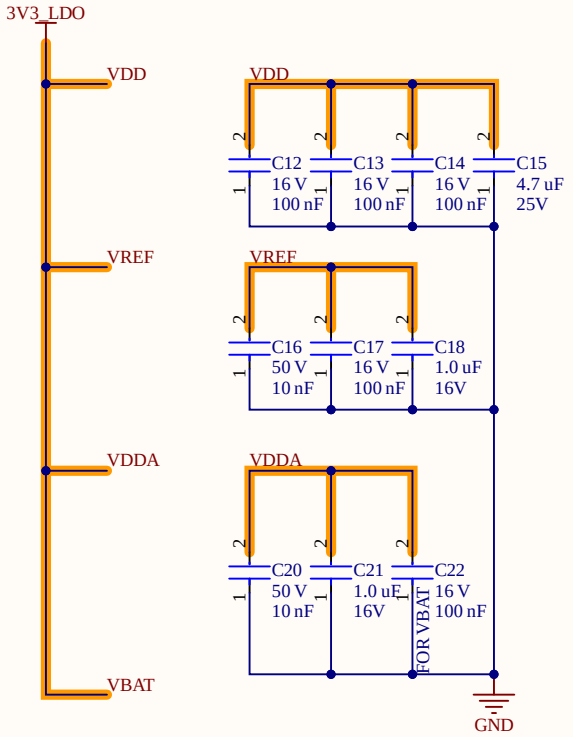
CAN USE IDRIVE FOR GATE SWITCHING NOISE - LIMITS FET SLEW RATE
REFERENCE: VESC
GATE RESISTORS FOR SWITCHING PULSES
PULL-DOWNS ON BOTTOM GATE TO AVOID FLOATING SHORTS
LOW PASS FILTERS ON SPA, SNA, FOR NOISE-FREE CURRENT READINGS

MCU

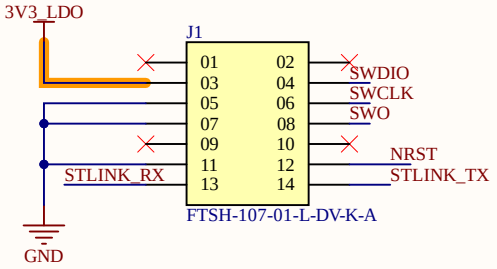


FOR PWM SIGNALS, USED HRTIM1_CHA1 HRTIM1_CHB1 HRTIM1_CHC1, AND CORRESPONDING CHA2, CHB2, CHC3

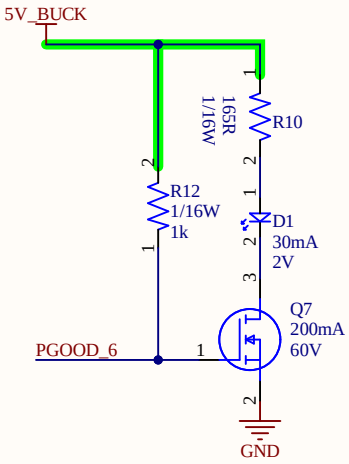
EXTRA FILTERING



SW FLASHING HEADER

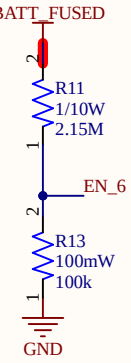


PGOOD LED



ENABLE

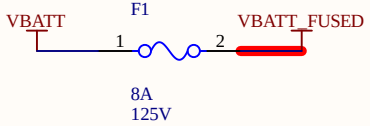
100k / (2.15M + 100k) * 48V = 2.13V
ABOVE ENABLE THRESHOLD



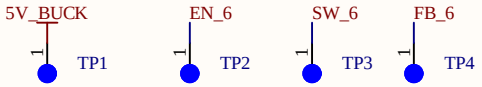
EXTRA FILTERING



FUSE



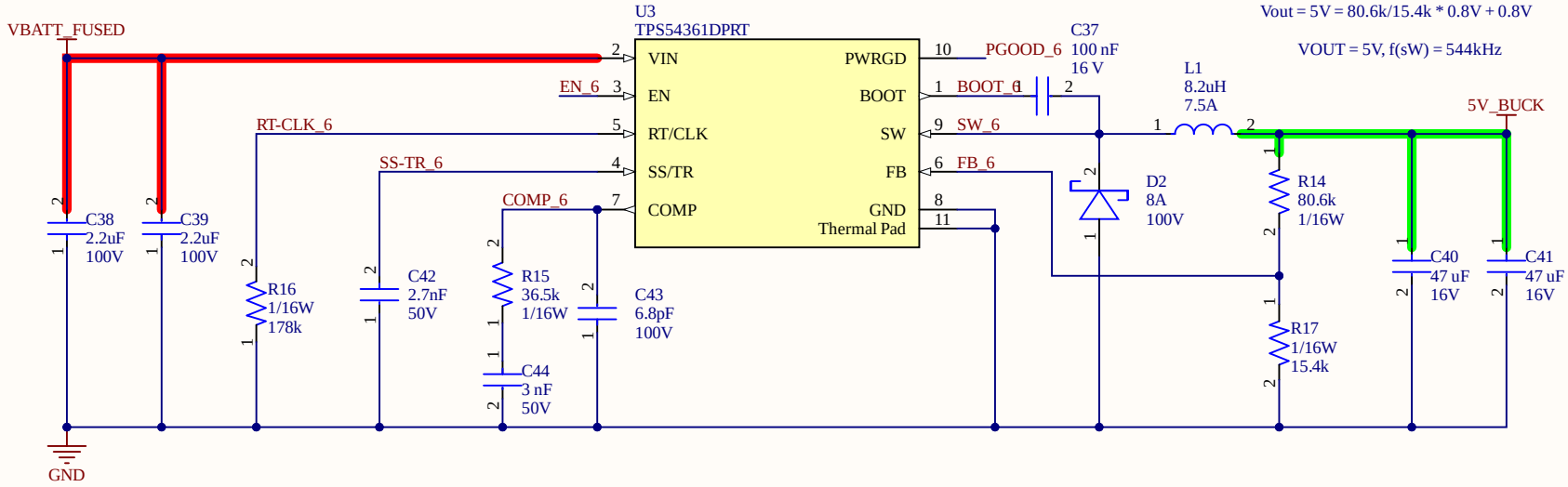
TEST POINTS



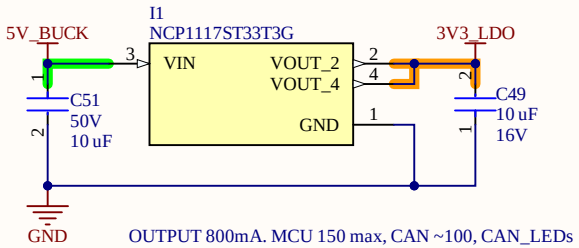
5V BUCK CONVERTER - LDO POWER

Vout = 5V = 80.6k/15.4k * 0.8V + 0.8V

VOUT = 5V, f(sW) = 544kHz

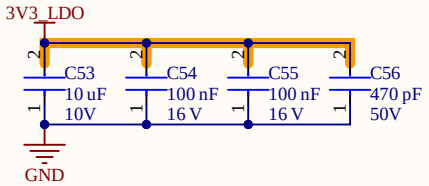


3V3 LDO

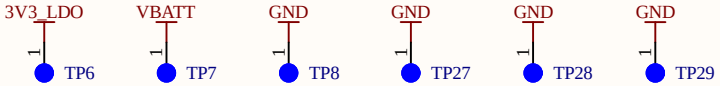


OUTPUT 800mA. MCU 150 max, CAN ~100, CAN_LEDs
DRIVES MCU, CAN, CAN_LEDS, ENCODER

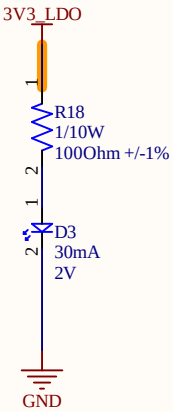
EXTRA FILTERING



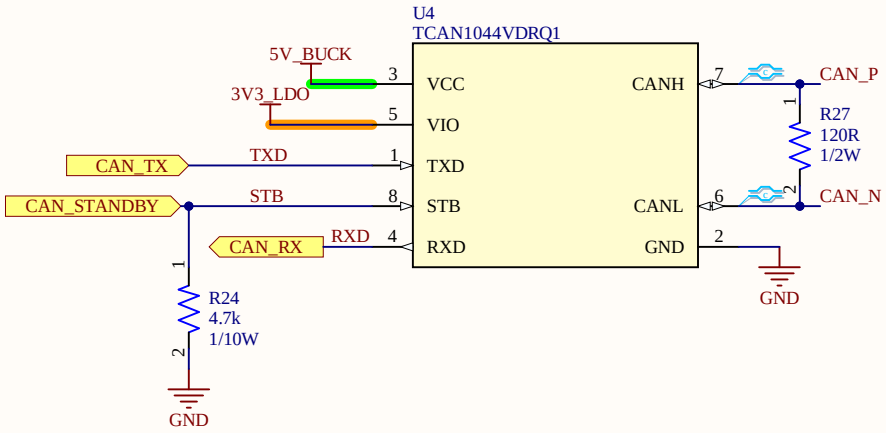
TEST POINTS



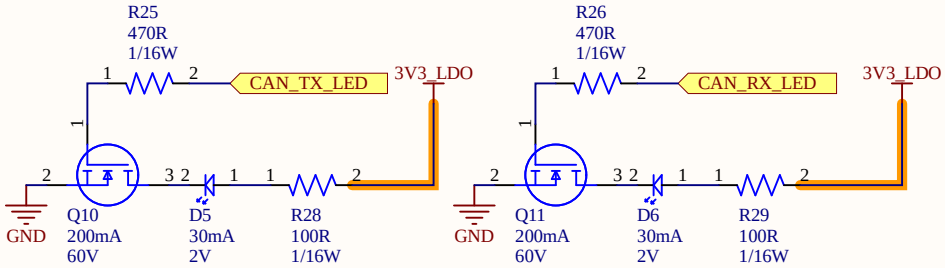
PGOOD 3V3



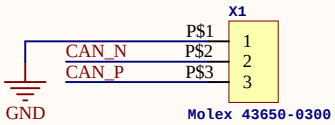
CAN TRANSCEIVER



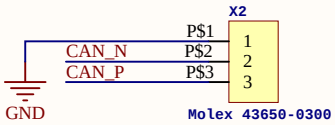
CAN LEDS



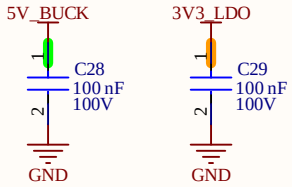
CAN_1



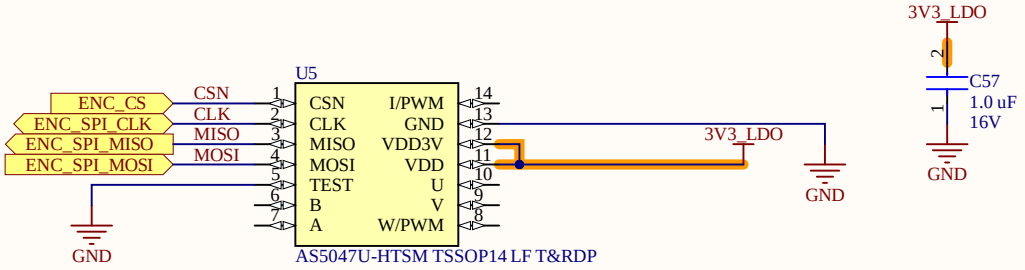
CAN_2



POWER FILTERING

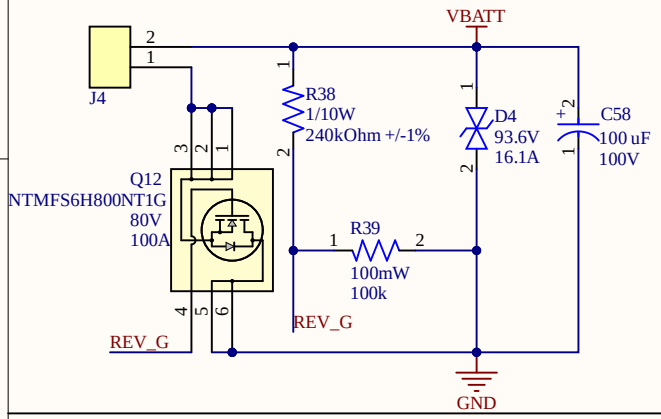


ABSOLUTE MAGNETIC ENCODER

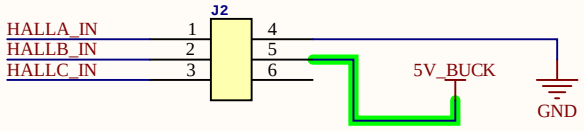


OTHER LINES FOR ABI/UVW COMMUNICATION PROTOCOL

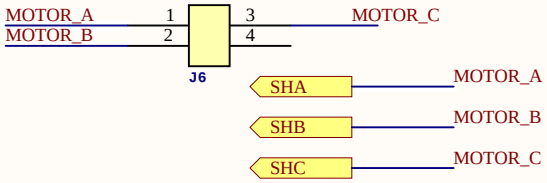
BATTERY REVERSE POLARITY



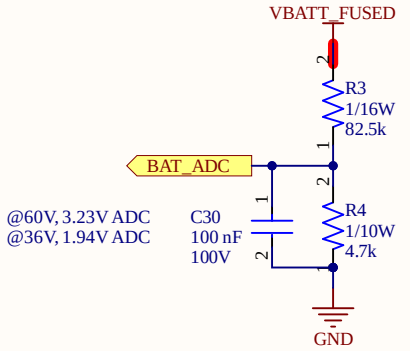
SENSOR CONNECTOR



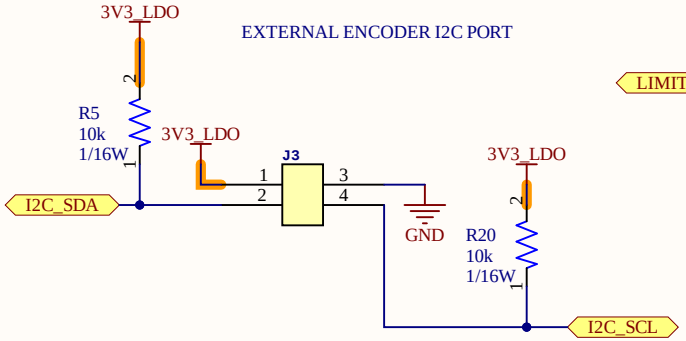
MOTOR CONNECTOR



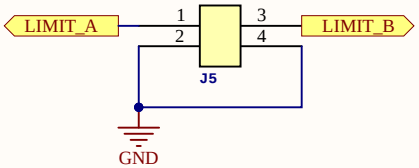
BATTERY VOLTAGE



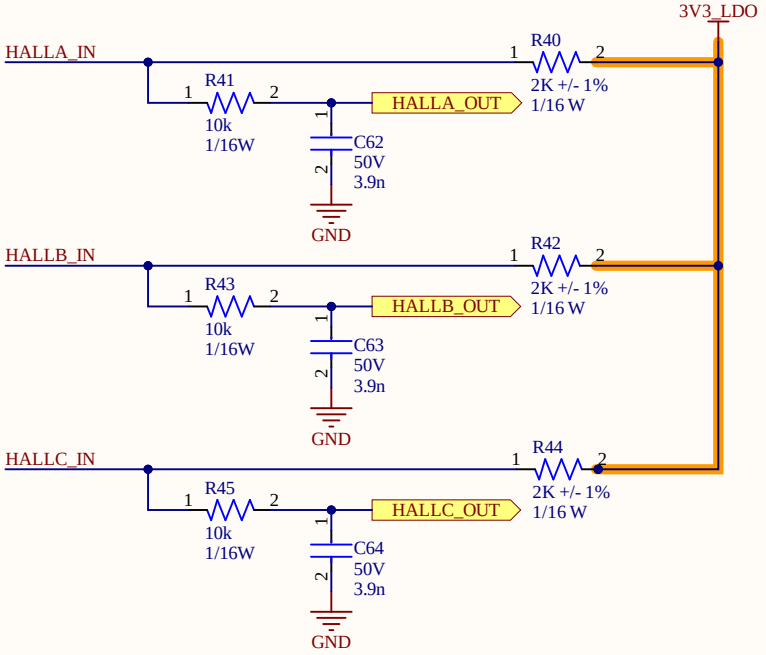
EXTERNAL ENCODER I2C PORT



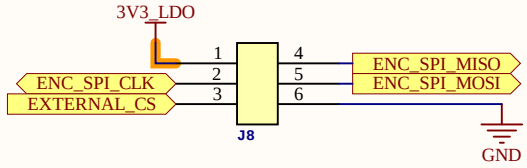
LIMIT SWITCHES



HALL FILTERING



EXTERNAL ENCODER SPI PORT



MOTOR TEMP CALCULATIONS:
TOTAL THERMAL RES: 2.06K/W
WINDING THERMAL TIME CONSTANT: 9.19s
30A W/ NO EMF: 218.5 C ABOVE STARTING
25 C TO 125 C IN 5.6s USING EXP TIME CONSTANT
LIMIT IN SOFTWARE FOR 2-3s